

ASHWIN SHARMA

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🎓 Bangalore Institute of Technology

PROFILE

Final-year Electronics and Communication Engineering student with hands-on experience in embedded C/C++, bare-metal firmware, and IoT systems on ARM and ESP32. Eager to grow into RTOS, Linux, and Android development through a hands-on embedded software internship.

EDUCATION

B.E. in Electronics and Communication Engineering | Bangalore Institute of Technology, Bengaluru | **CGPA: 8.64** | 2023–2027

Relevant Coursework: VLSI Design, Digital Electronics, Microarchitecture, Computer Networks, Embedded Systems.

SKILLS and TOOLS

Category	Skills / Tools
Hardware Description Languages	Verilog
Programming Languages	Embedded C, C Programming, Python
Networking & Connectivity	UDP sockets, TCP/IP, WiFi (ESP32), OSI Model, playit.gg tunneling, JSON over UDP
EDA / VLSI Tools	Cadence Genus (Synthesis), Cadence Xcelium (Simulation), Icarus Verilog (Functional Verification), GTKWave (Simulation/Waveform Analysis), Xilinx Vivado (Spartan-3)
Development Tools & Platforms	Git / GitHub, Keil uVision, Wokwi Simulator, Tinkercad, Arduino IDE

PROJECTS

MICROKERNAL-F1 | Bare Metal C | ARM Cortex-M3 |

- Built a bare-metal RTOS-style scheduler on STM32F103 from scratch in C (no HAL/CubeMX), custom PLL clock configuration, interrupt-driven SysTick heartbeat, non-blocking USART driver with shell-ready ring-buffered I/O using dual ring buffers with race-condition-safe critical sections.
- Debugged and resolved low-level embedded issues independently, including linker symbol conflicts, clock-tree misconfigurations, and ISR/main-loop synchronization hazards, through systematic register-level analysis. [open-sourced](#)

IoT Network Pipeline - Environmental Monitoring System | Embedded C | Python | ESP32(with WIFI) | UDP |

- Built an ESP32-based IoT environmental monitoring system integrating DHT22 and MQ2 sensors with real-time data streaming over UDP sockets via WiFi, tunneled through playit.gg to a Python base station for live alerting and CSV logging.
- Simulated and validated the full networking pipeline on Wokwi, applying core networking concepts including Client-Server architecture, UDP/TCP protocols, and socket programming across OSI layers 3 and 4 [open-sourced](#)

Arduino LED Tester + Resistor Calculator | Embedded C | PWM | I2C | Analog Signal Processing |

- Built a bench instrument, measuring LED forward voltage and actual current via a 10Ω sense resistor with a closed-loop PWM control algorithm regulating test current (1-20 mA), displayed on an I2C 16×2 LCD
- Implemented a lookup engine mapping calculated resistance to the nearest E24 standard value, with power-dissipation safety warnings (>0.5 W) and full serial monitor logging. [open-sourced](#)

WORKSHOPS and CERTIFICATIONS

- RTL-to-GDSII Flow on SRAM (5 Days):** Cadence Xcelium, Genus, Modus, Innovus: Learned to build an S-RAM cell from RTL netlist to verified GDS layout. ECE Dept., BIT × VLSIMINDS | 2026
- SEMulator3D Physical Design Workshop (3 Days):** SEMulator 3D : Learned to build Transistor & FinFET structures along with process variability and device scaling. IISc Bengaluru × IEEE × Lam Research | 2025
- IBM SkillsBuild – Getting Started with Artificial Intelligence** IBM | Mar 2026 | Verified on Credly

HACKATHONS & COMPETITIONS

- Cisco Next Gen Champions League 2026:** AI-driven predictive hardware health monitor using edge ML on ESP32/STM32 with multi-sensor fusion | Round 1 cleared
- Xcelerate 24hr Hackathon, NIKSHATRA E-Summit 2025:** MINDCARE+, a student mental health monitoring app | National level, Team Nexus